

PULINDU PASANJITH

Software Engineering Undergraduate

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[GitHub](#) | [LinkedIn](#) | [Portfolio Website](#)

SUMMARY

Software engineering undergraduate building production-style AI systems end-to-end, without relying on high-level frameworks, to demonstrate a genuine working understanding of the underlying mechanics. Built ResolveAI, a Retrieval-Augmented Generation assistant covering embeddings, vector search, and grounded generation, and InsightForge, an autonomous research agent that plans its own searches, reads sources, and self-corrects when its tool-calling drifts. Comfortable across front-end interfaces, Python-based ML/NLP and agentic pipelines, REST API design, and collaborative version-controlled development. Brings a structured, detail-oriented approach from 6+ years managing operations and reporting systems, paired with active coursework in software architecture, requirements engineering, and data structures. Seeking a Software Engineer, AI Engineer, or Junior Developer role to apply and grow these skills in a professional team.

PROJECTS

ResolveAI — RAG-Based Customer Support Assistant *(Python, FastAPI, ChromaDB, React, Gemini API, Docker)*

github.com/pulindu117/ResolveAI

- Designed and built a full-stack Retrieval-Augmented Generation system from first principles (no LangChain), implementing document ingestion, character-based chunking with overlap, embedding generation, and semantic retrieval as independent, reusable modules.
- Implemented semantic search using BAAI/bge-small-en-v1.5 sentence embeddings and ChromaDB for persistent vector storage, retrieving top-k relevant chunks via cosine similarity to ground LLM responses and reduce hallucination.
- Built a REST API with FastAPI (chat, document upload, list, and delete endpoints) backed by a pluggable LLM provider architecture, and a React + Tailwind CSS frontend with a citation-aware chat interface.
- Containerized the full stack with Docker and Docker Compose for one-command deployment, and wrote automated pytest test coverage for the chunking pipeline.

InsightForge — Autonomous AI Research Agent *(Python, Gemini API, Tavily, pytest)*

github.com/pulindu117/InsightForge

- Built an autonomous research agent from raw Gemini API calls (no agent framework) that plans its own search queries, retrieves and reads web sources via tool calling, and iteratively decides whether it has sufficient evidence before producing a cited Markdown report.
- Designed a self-correction loop that detects when the model deviates from expected tool-calling behaviour, issues a corrective re-prompt, and falls back to capturing the model's output directly if correction fails — preventing completed research from being silently discarded.
- Implemented an input sanitisation layer that validates and coerces LLM-generated tool arguments against a schema whitelist, protecting the tool execution layer from malformed or hallucinated function-call parameters.
- Wrote an 11-test pytest suite covering tool behaviour, network failure handling, and input sanitisation using mocked external calls, allowing the full suite to run without API keys or network access.

Personal Portfolio Website *(HTML, CSS, JavaScript)*

github.com/pulindu117/portfolio-website

- Designed and built a responsive personal portfolio site from scratch using HTML, CSS, and JavaScript, then deployed it via GitHub.
- Implemented a contact/intake form that writes submitted visitor data directly into a connected Google Sheet, and added a one-click CV download feature.

Fake News Detection (NLP Group Project) *(Python, scikit-learn, TensorFlow/Keras, Pandas)*

github.com/pulindu117/NLP_Group_04

- Group NLP project (team of 3, role: project lead) building a fake news classifier on the WELFake dataset (72,000+ news articles) using text preprocessing, feature extraction, and supervised learning.
- Built and evaluated a Logistic Regression baseline and an LSTM deep learning model for binary text classification using a structured, version-controlled notebook pipeline.

PROFESSIONAL EXPERIENCE

Operations & IT Executive — Deepna Transport Pvt. Ltd.

Colombo, Sri Lanka

2019 – Present

- Built Excel-based systems to track trips, fuel usage, income, and detention charges, cutting calculation errors by 90% and improving reporting turnaround by 80%.
- Managed monthly billing, invoicing, payroll, and compliance documentation for container transport operations, working with management to analyze income trends and cost-saving opportunities.

Open Source Contributor — GitHub (First Contributions)

2025

- Completed a full open-source contribution workflow (fork, branch, commit, pull request) with a PR merged into the official repository — practicing collaborative version control as used in professional engineering teams.

EDUCATION

Bachelor of Software Engineering (Honours) — SLTC Research University

2024 – 2028

Padukka, Sri Lanka | GPA: 3.7

- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Software Architecture & Design, Software Quality Assurance, Requirements Engineering, Natural Language Processing.

Level 3 Diploma in Management (UK) — ESOF Metro Campus, 2024–2025 (8 Distinctions, 2 Merits)

Assured Diploma in Information Technology (DITEC) — Pearson Assured, 2023–2024 (Distinction)

Assured Diploma in English (DIE) — Pearson Assured, 2023–2024 (Distinction)

GCE Advanced Level Examination — Royal College, Colombo 7, 2018–2020 (Mathematics Stream)

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML/CSS, Bash

AI / Agentic Engineering: Retrieval-Augmented Generation (RAG), autonomous tool-calling agents, sentence embeddings, vector databases (ChromaDB), semantic search, prompt engineering, LLM API integration (Anthropic Claude, Gemini)

Backend & Tools: FastAPI, REST API design, SQLite, Docker, Git/GitHub, pytest

Frontend: React, Vite, Tailwind CSS

Machine Learning / Data: ML fundamentals, neural networks, NLP, NumPy, Pandas, Matplotlib, scikit-learn, Google Colab/Jupyter, Vertex AI (Google Cloud)

Other: Cisco Packet Tracer (network simulation & troubleshooting), Advanced Excel/Google Sheets

CERTIFICATIONS & LANGUAGES

Google Cloud — Generative AI Learning Path (2025) | AWS — Fundamentals of ML & AI (2025) | Cisco — Intro to Modern AI & Packet Tracer (2025)

Languages: Sinhala (Native), English (Proficient), Tamil (Intermediate)